

A

B

C

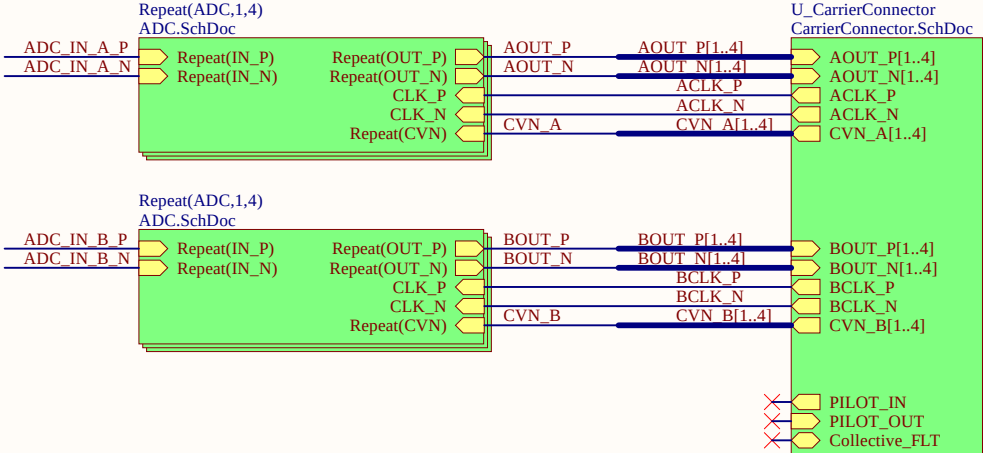
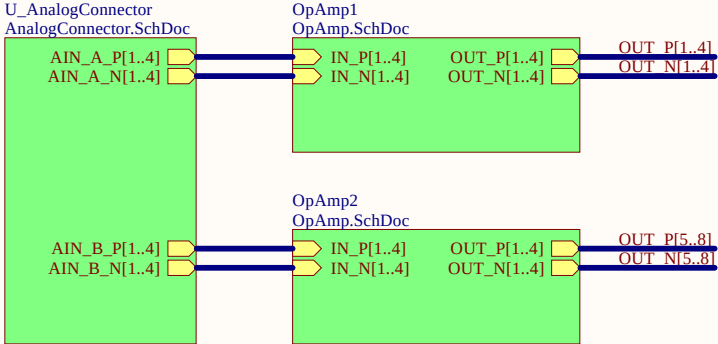
D

A

B

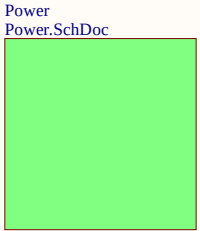
C

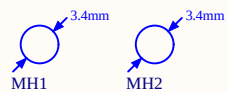
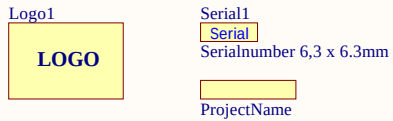
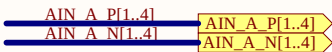
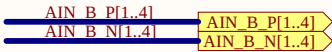
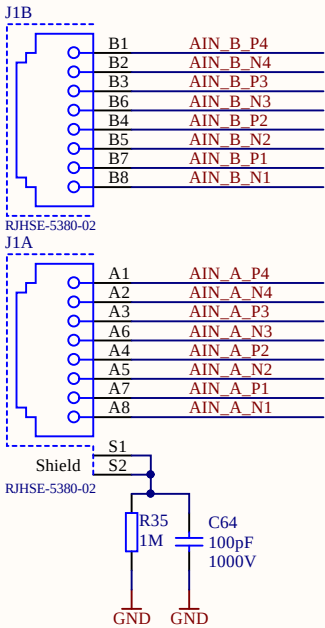
D



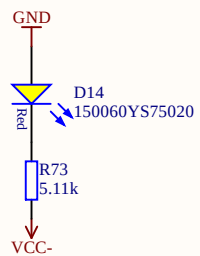
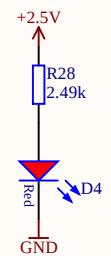
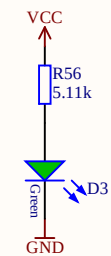
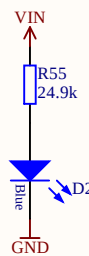
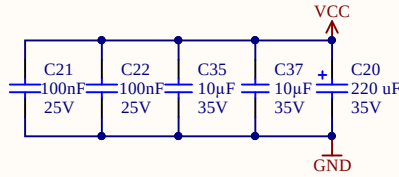
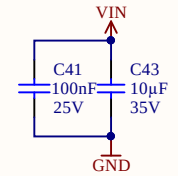
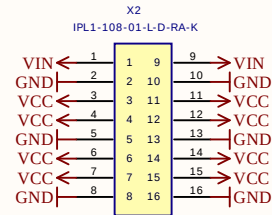
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OUT_N1	ADC_IN_A_N1
OUT_P2	ADC_IN_A_P2
OUT_N2	ADC_IN_A_N2
OUT_P3	ADC_IN_A_P3
OUT_N3	ADC_IN_A_N3
OUT_P4	ADC_IN_A_P4
OUT_N4	ADC_IN_A_N4

OUT_P5	ADC_IN_B_P1
OUT_N5	ADC_IN_B_N1
OUT_P6	ADC_IN_B_P2
OUT_N6	ADC_IN_B_N2
OUT_P7	ADC_IN_B_P3
OUT_N7	ADC_IN_B_N3
OUT_P8	ADC_IN_B_P4
OUT_N8	ADC_IN_B_N4



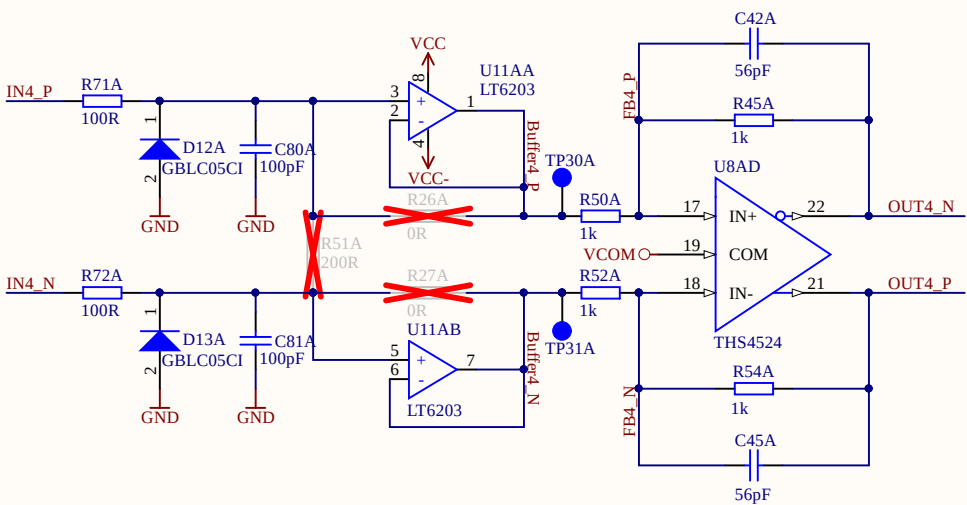
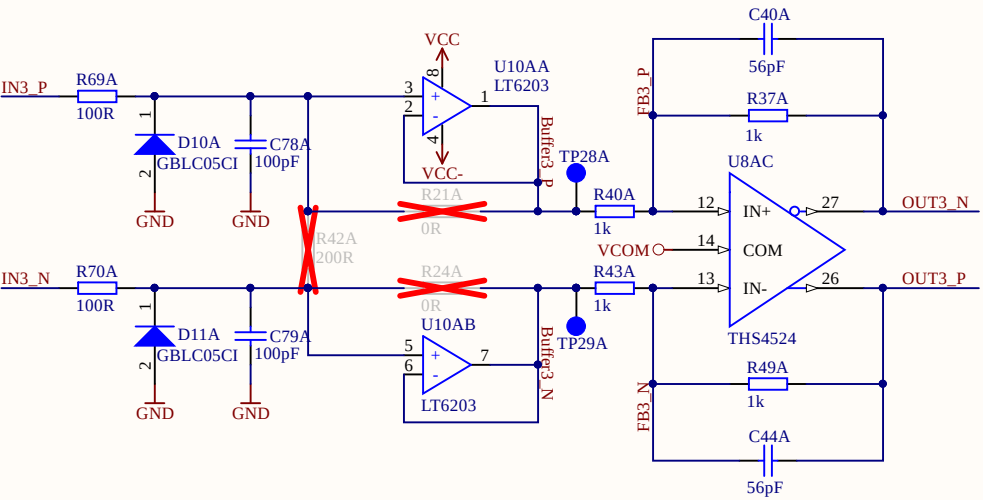
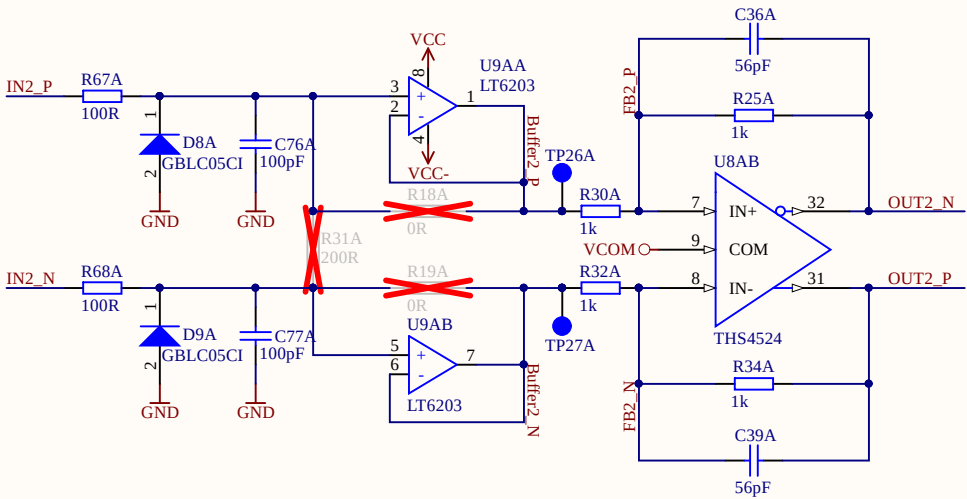
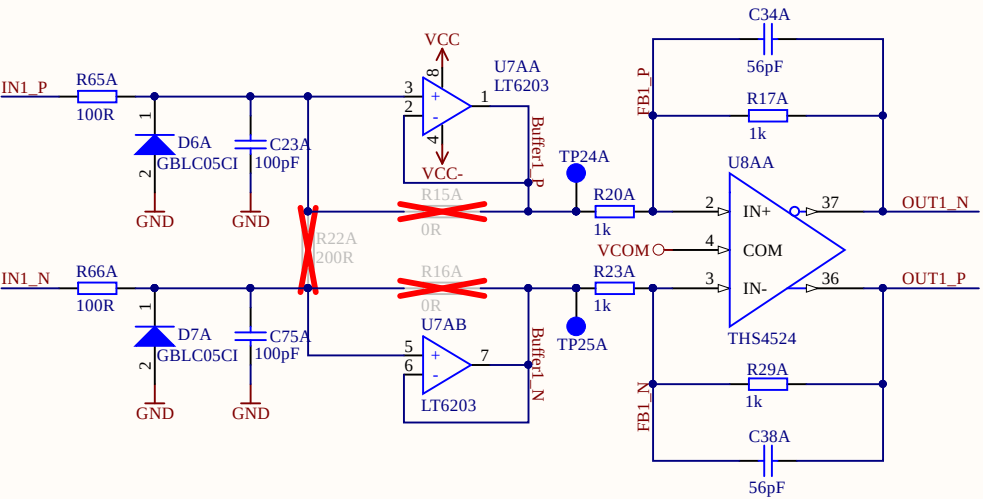


When using the VIN on the power connector X1, use pin 2 and 10 as return ground



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Size: A4	Revision: 3v02			
Date: 27.07.2020	Time: 09:57:14	Sheet 2 of 14	Author: Thomas Kreppel	
Project: UltraZohm_ADC.PrjPCB				

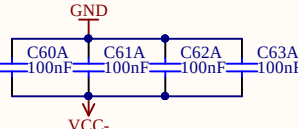
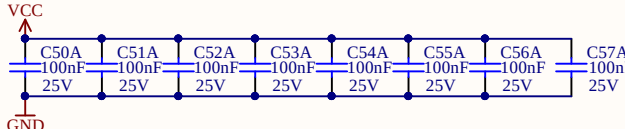
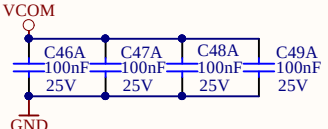
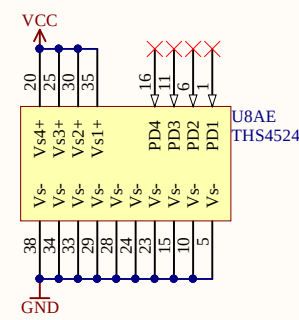
Without buffer: fit R15,R16 with 0R; U7 dnp
With buffer: R15,R16 dnp ; U7 fitted
Current signal: R15,R16 dnp ; U7 fitted; R22 measurement shunt



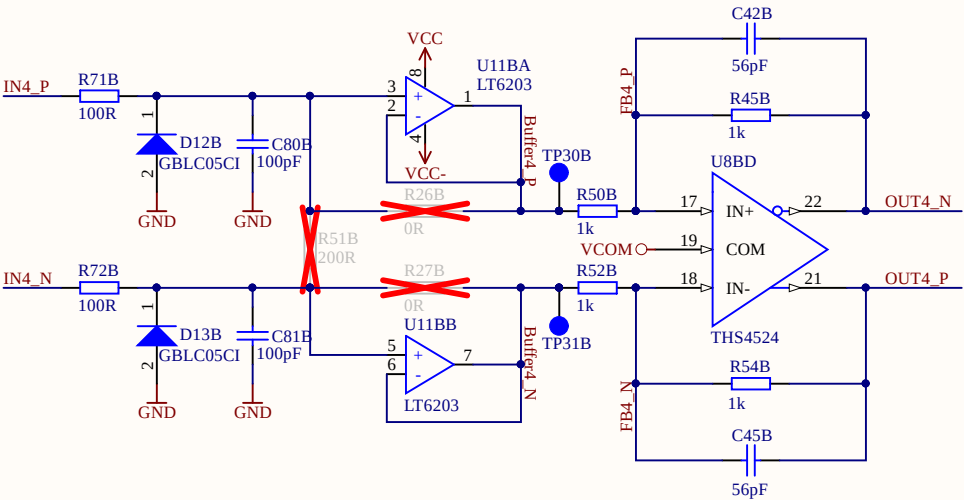
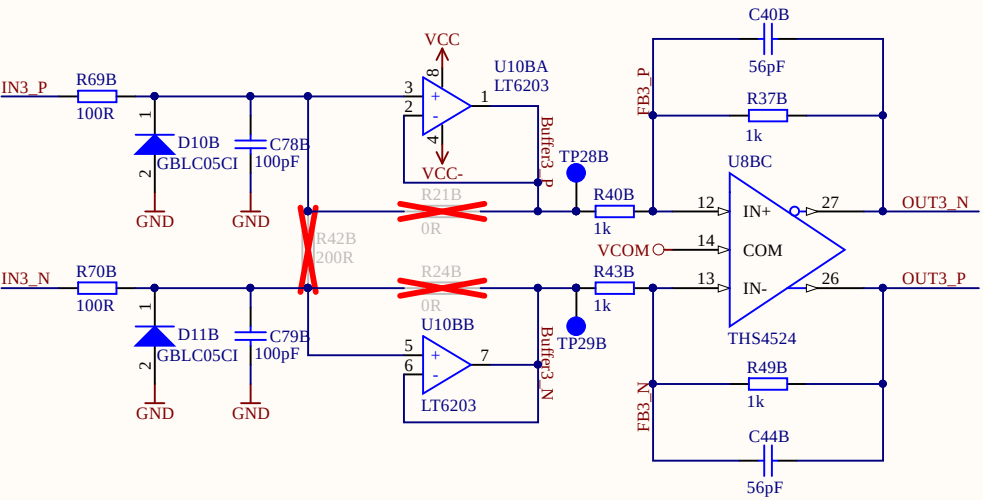
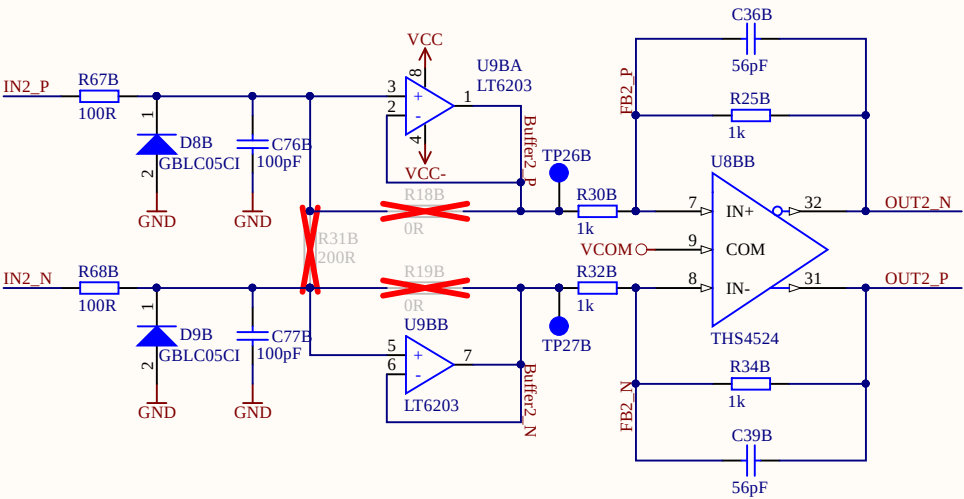
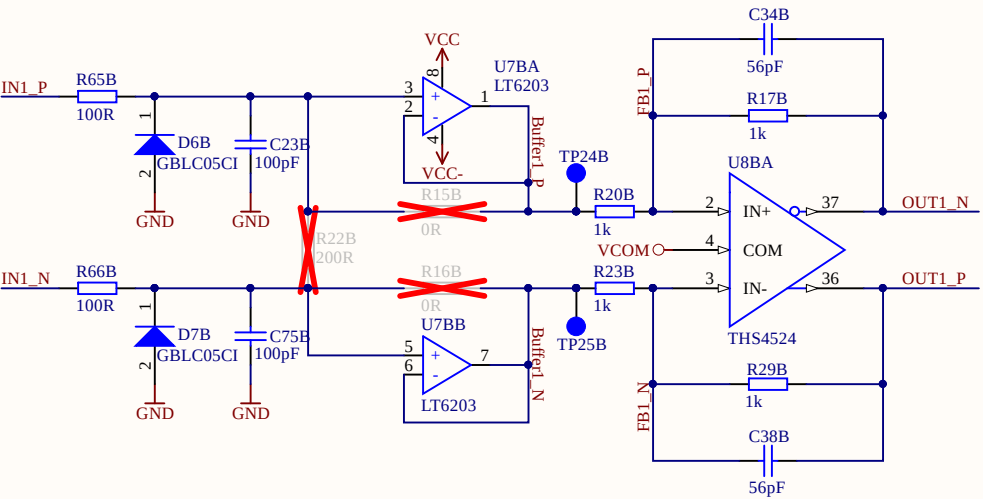
IN_P[1..4] IN_N[1..4]

OUT_P[1..4] OUT_N[1..4]

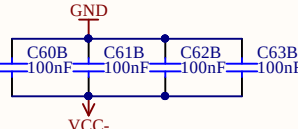
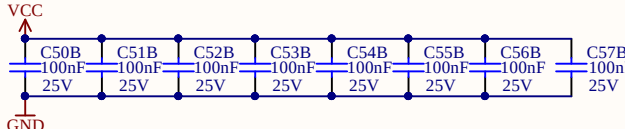
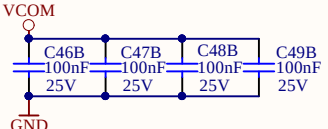
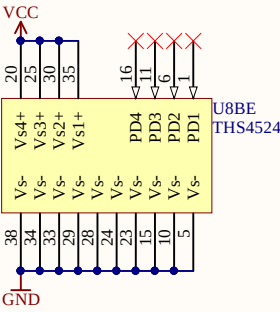
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IN1_N	IN_N1
IN2_P	IN_P2
IN2_N	IN_N2
IN3_P	IN_P3
IN3_N	IN_N3
IN4_P	IN_P4
IN4_N	IN_N4
OUT1_P	OUT_P1
OUT1_N	OUT_N1
OUT2_P	OUT_P2
OUT2_N	OUT_N2
OUT3_P	OUT_P3
OUT3_N	OUT_N3
OUT4_P	OUT_P4
OUT4_N	OUT_N4



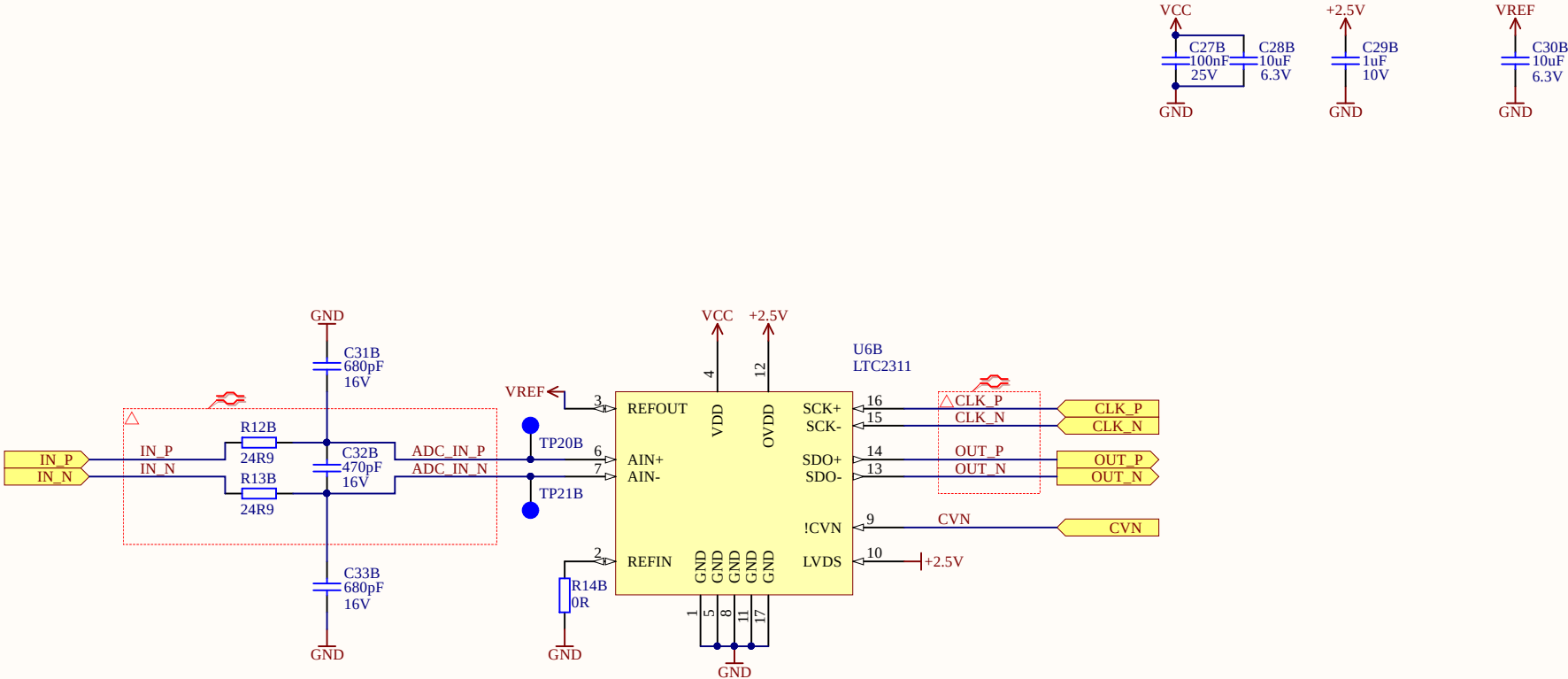
Without buffer: fit R15,R16 with 0R; U7 dnp
With buffer: R15,R16 dnp ; U7 fitted
Current signal: R15,R16 dnp ; U7 fitted; R22 measurement shunt



IN_P[1..4]	IN_P[1..4]
IN_N[1..4]	IN_N[1..4]
OUT_P[1..4]	OUT_P[1..4]
OUT_N[1..4]	OUT_N[1..4]
IN1_P	IN_P1
IN1_N	IN_N1
IN2_P	IN_P2
IN2_N	IN_N2
IN3_P	IN_P3
IN3_N	IN_N3
IN4_P	IN_P4
IN4_N	IN_N4
OUT1_P	OUT_P1
OUT1_N	OUT_N1
OUT2_P	OUT_P2
OUT2_N	OUT_N2
OUT3_P	OUT_P3
OUT3_N	OUT_N3
OUT4_P	OUT_P4
OUT4_N	OUT_N4
FB1_P	
FB1_N	
FB2_P	
FB2_N	
FB3_P	
FB3_N	
FB4_P	
FB4_N	
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Buffer4_P	
Buffer1_N	
Buffer2_N	
Buffer3_N	
Buffer4_N	

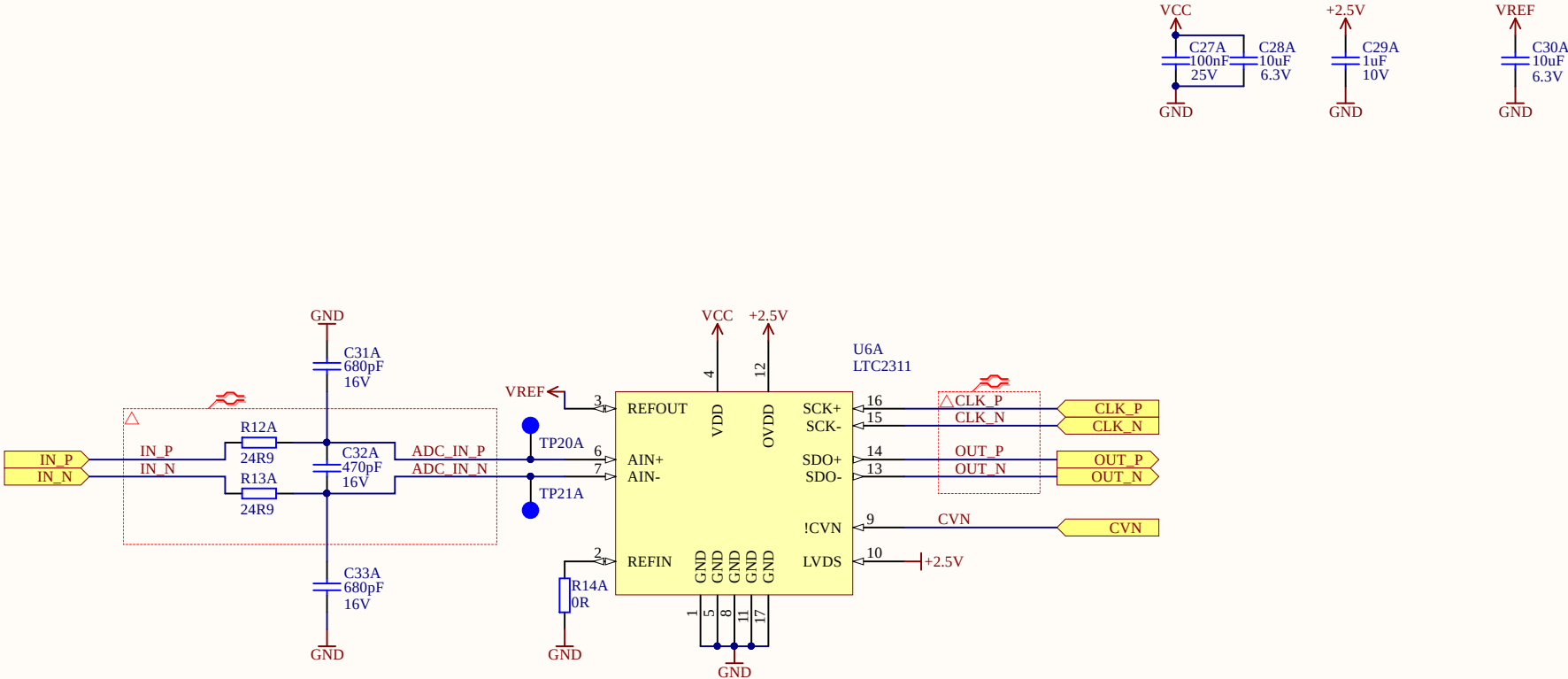


ADC can provide internal 4.096V reference. For that purpose R14 has to be replaced by 10u capacitor



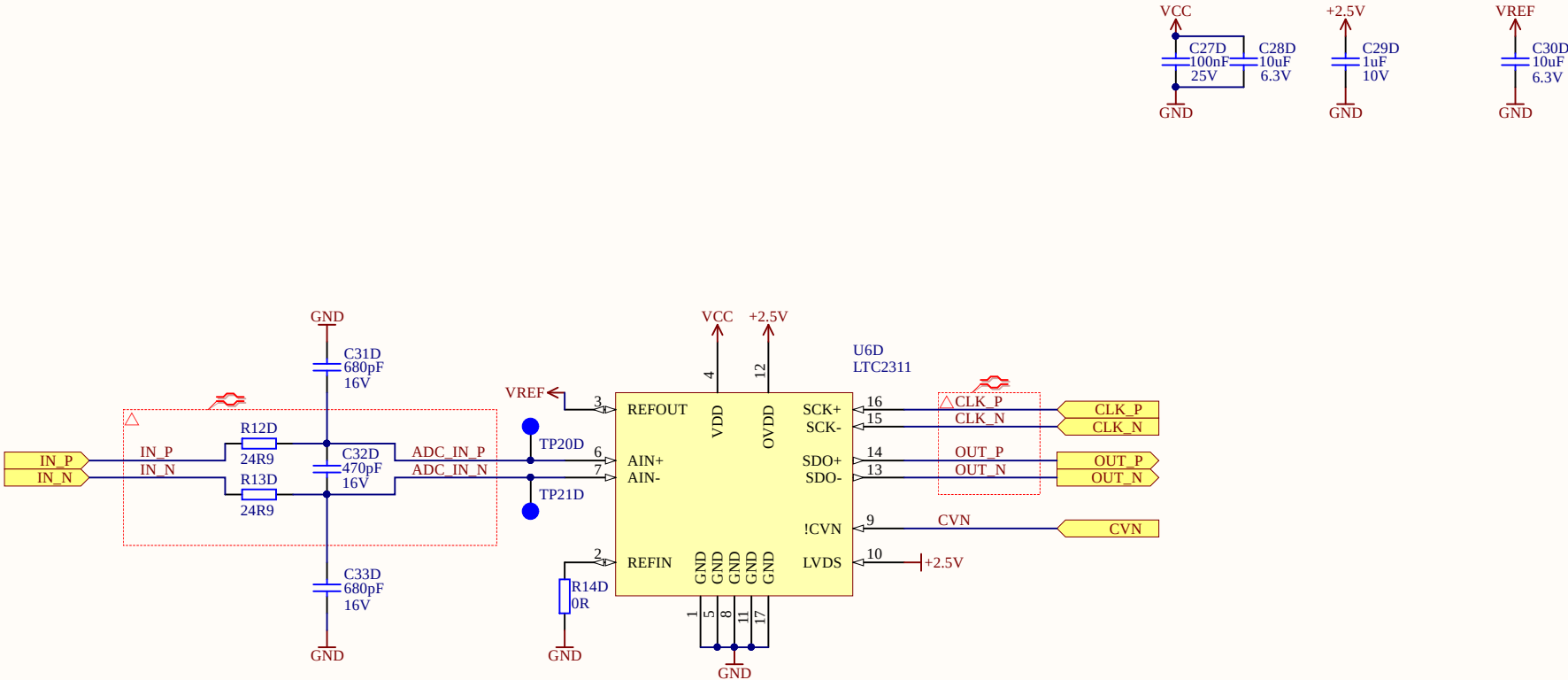
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Date: 27.07.2020	Time: 09:57:15	Sheet 4.1 of 14	Author: Thomas Kreppel
Project: UltraZohm_ADC.PrjPCB			

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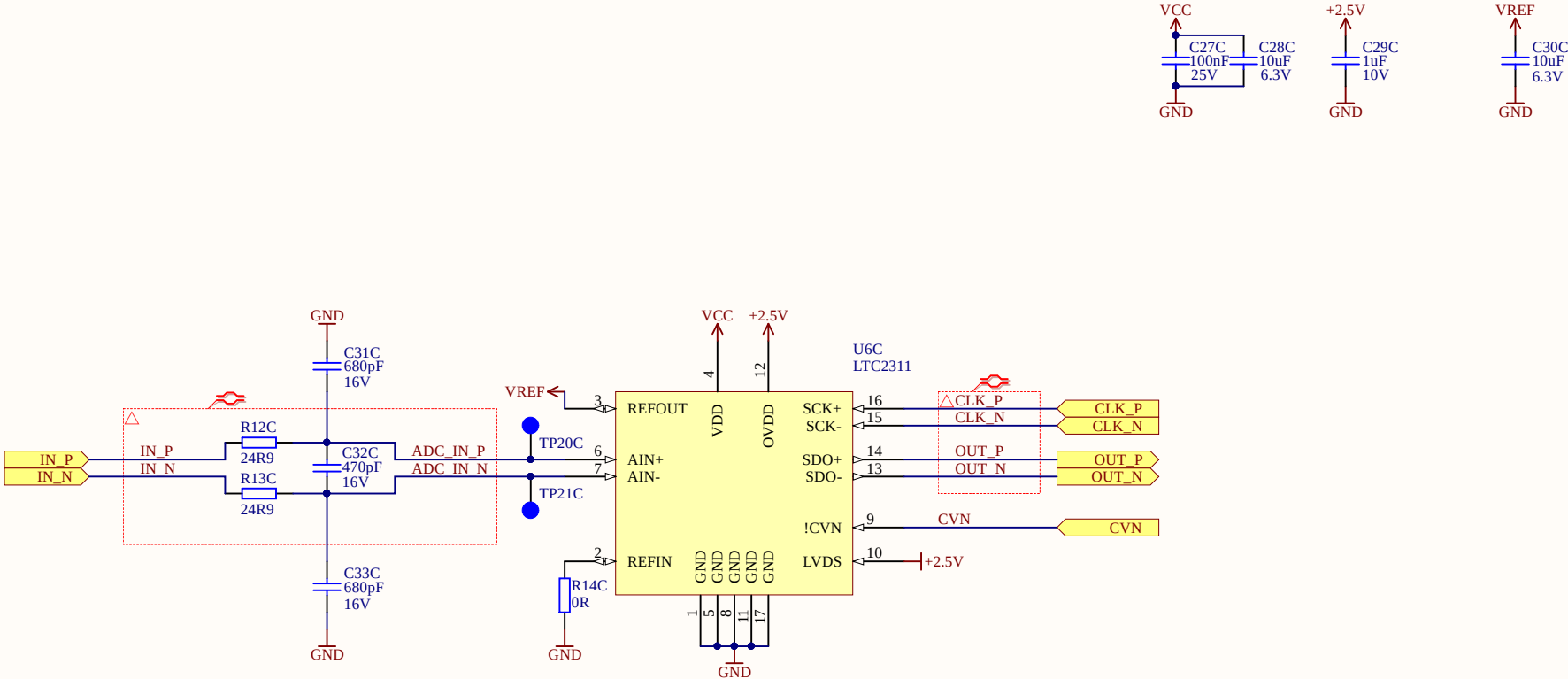
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Size: A4	Revision: 3v02			
Date: 27.07.2020	Time: 09:57:15	Sheet 4.1 of 14	Author: Thomas Kreppel	
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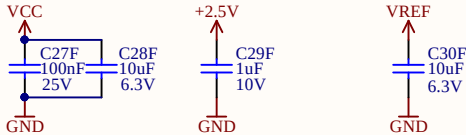
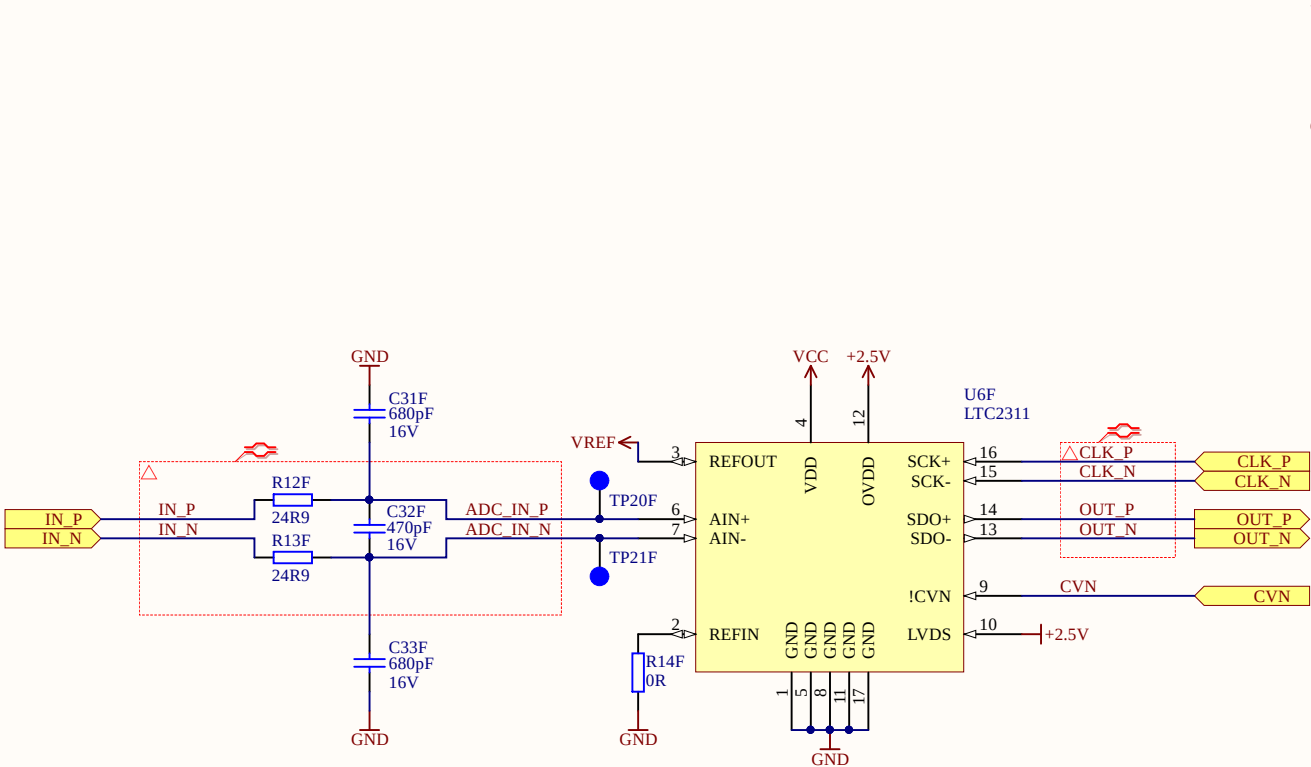
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Date: 27.07.2020	Time: 09:57:15	Sheet 4.2 of 14	Author: Thomas Kreppel
Project: UltraZohm_ADC.PrjPCB			

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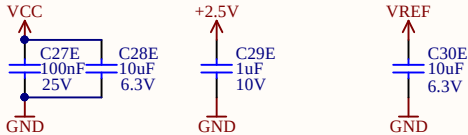
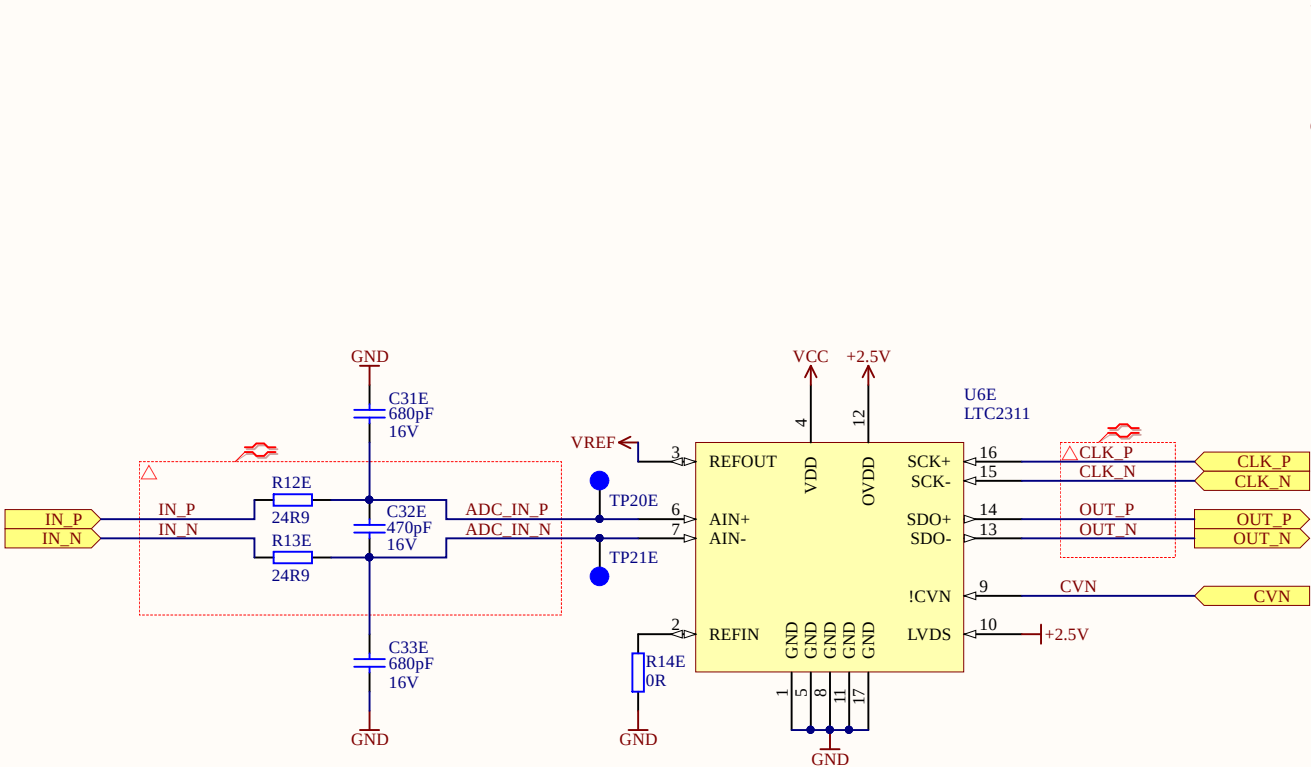
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Date: 27.07.2020	Time: 09:57:15	Sheet 4.2 of 14	Author: Thomas Kreppel
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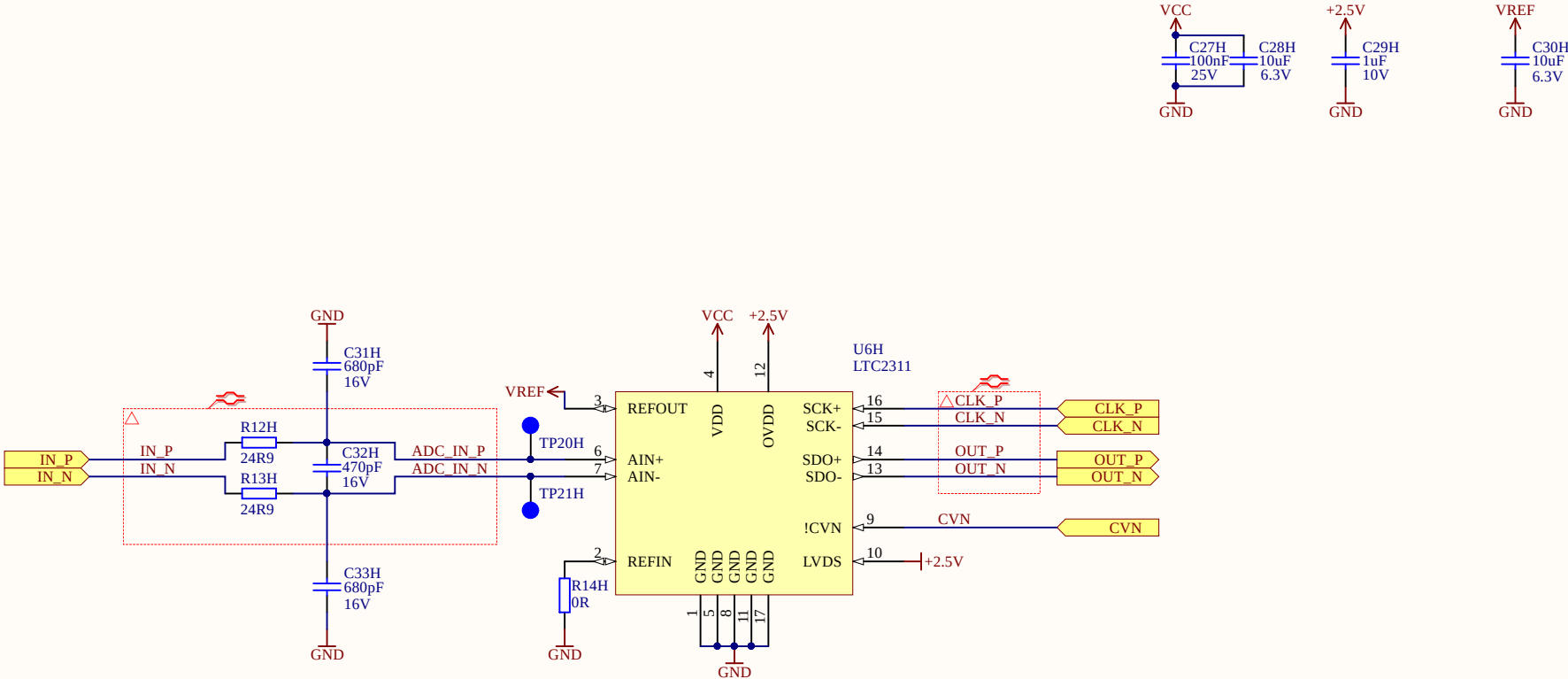
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Size: A4	Revision: 3v02	Sheet 4.3 of 14		Author: Thomas Kreppel
Date: 27.07.2020	Time: 09:57:15	Project: UltraZohm_ADC.PrjPCB		

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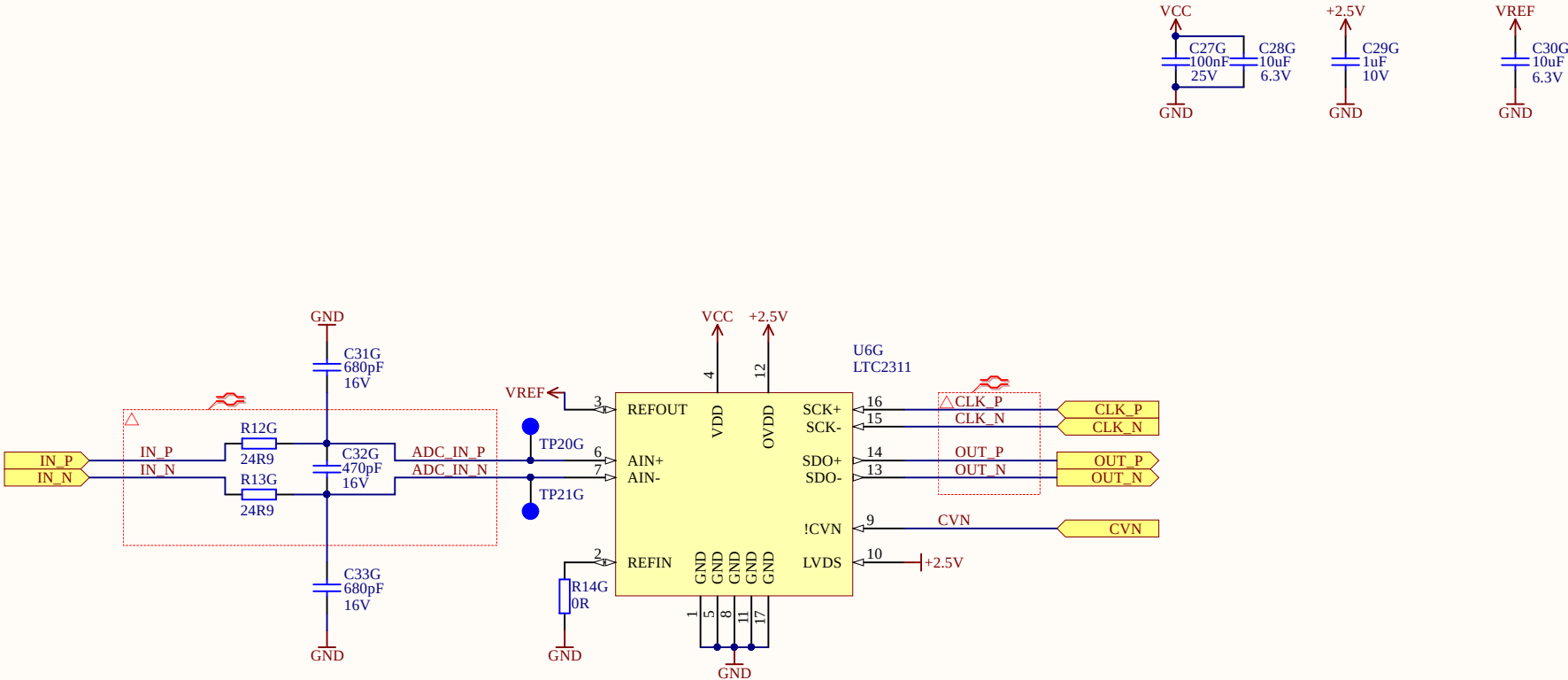
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Size: A4	Revision: 3v02	Sheet 4.3 of 14		Author: Thomas Kreppel
Date: 27.07.2020	Time: 09:57:15	Project: UltraZohm_ADC.PrjPCB		

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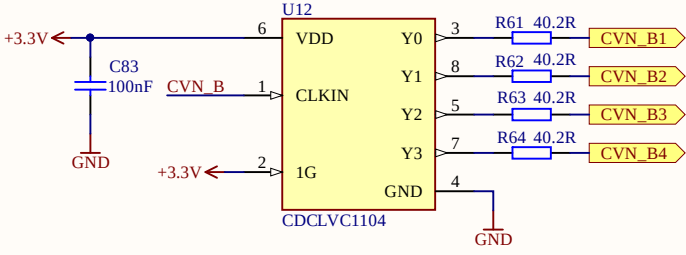
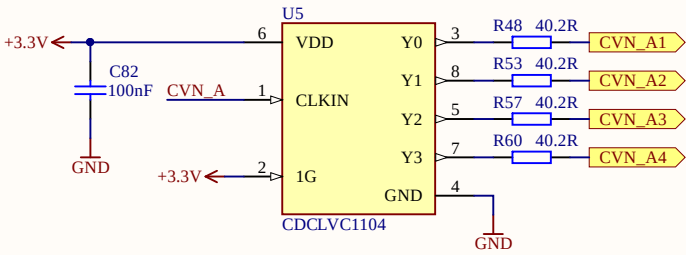
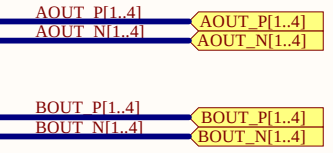
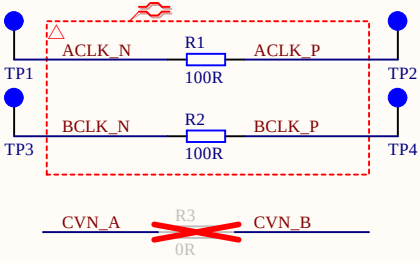
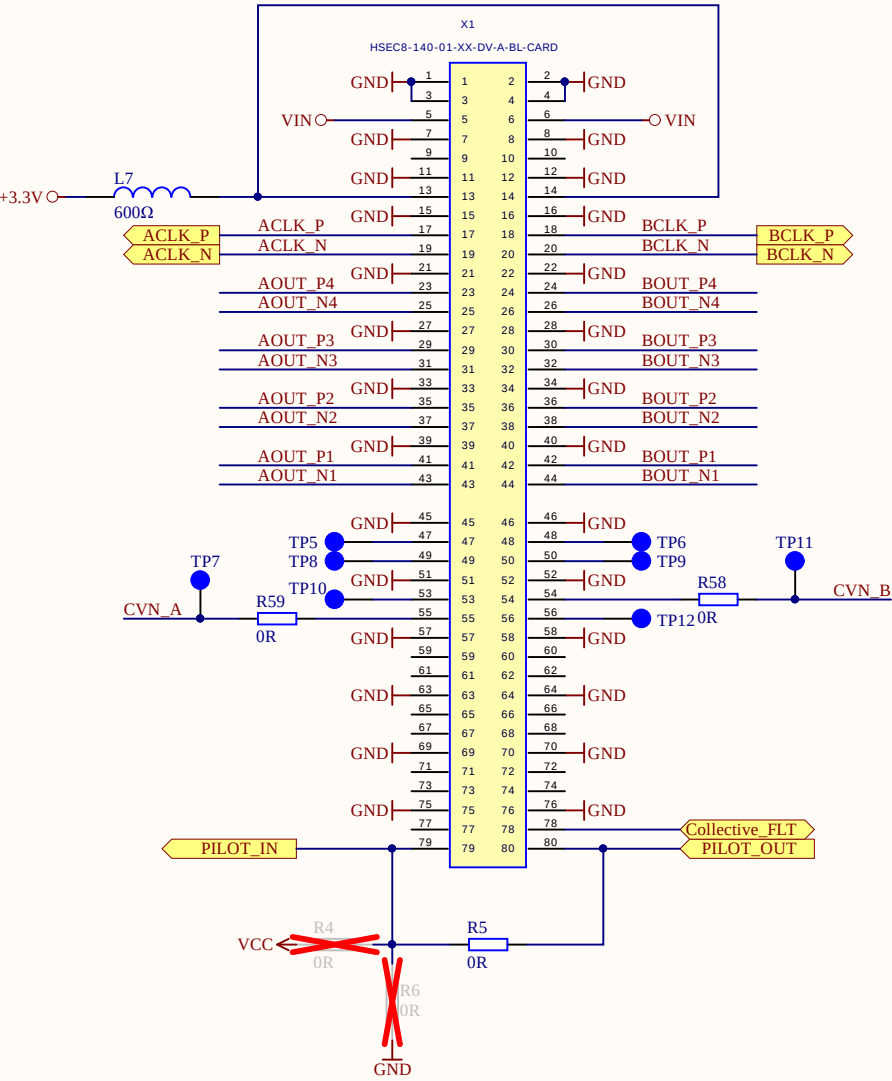
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Date: 27.07.2020	Time: 09:57:15	Sheet 4.4 of 14	Author: Thomas Kreppel
Project: UltraZohm_ADC.PrjPCB			

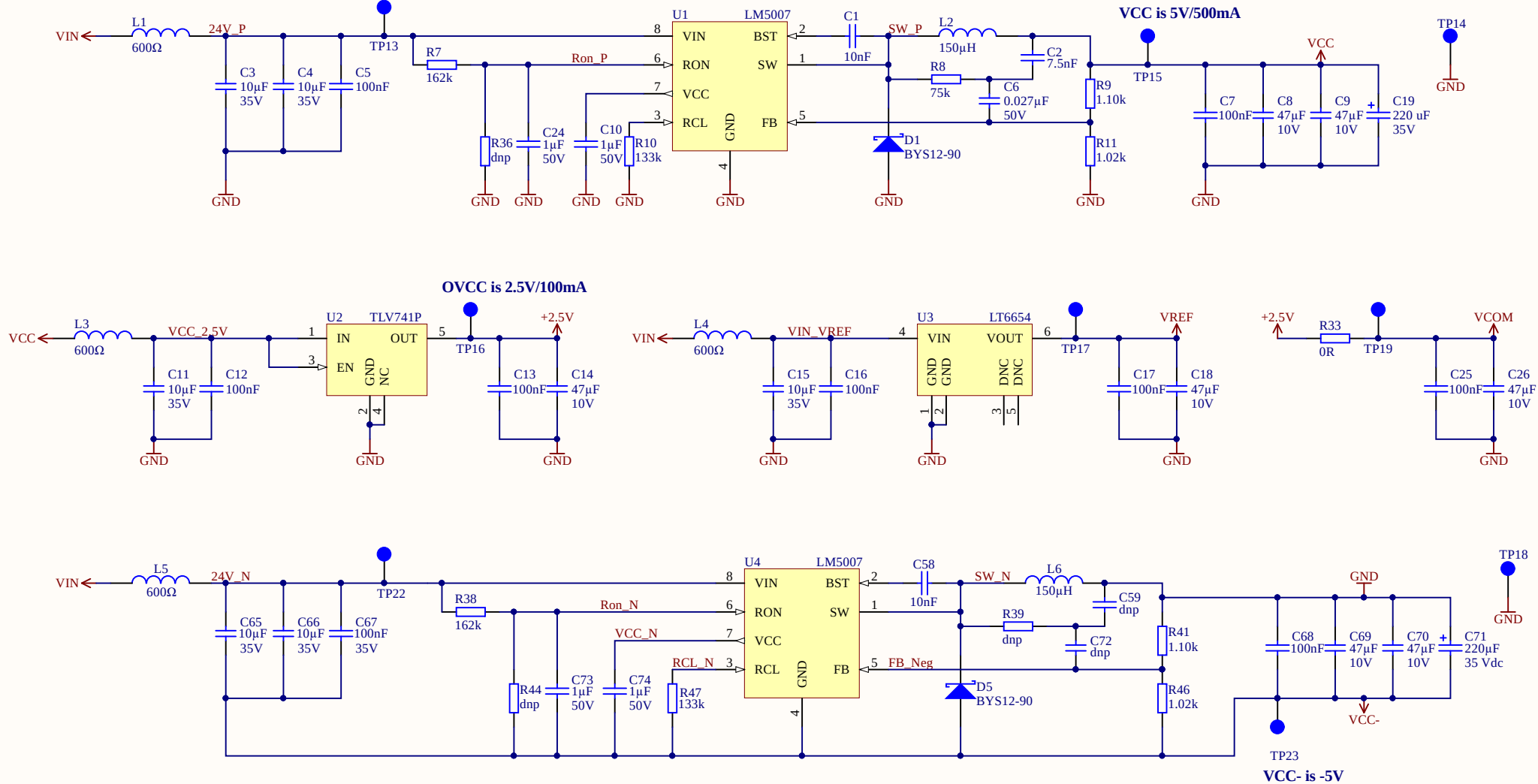
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Size: A4	Revision: 3v02			
Date: 27.07.2020	Time: 09:57:16	Sheet 4.4 of 14	Author: Thomas Kreppel	
Project: UltraZohm_ADC.PrjPCB				

If R60 is fitted with 0R all ADCs are controlled with one conversion signal. R61 and R62 are then dnp
R50 and R51 (R61 and R62) are used to adjust 3.3V CVN signal to 2.5V level
R52, R53 and R54 are fitted appropriately for Pilot In/Out





LT6655 is available in 1.25V/2.048V/2.5V/3V/3.3V/4.096V/5V
If the internal reference from LTC2311 is used U3 must not be fitted

For +/-5.0V rails use 1kOhm resistors (CRCW06031K00JNEAC) for R6, R8, R12, R14

Title Power.SchDoc		EAL TU München Arcisstraße 21 80333 München		\\nas.ads.mwn.de\ga42ka f\dokumente\Altium\AD C_Board\Logo_TUM2.p ng	
Size: A4	Revision: 3v02				
Date: 27.07.2020	Time: 09:57:16	Sheet 6	of 14	Author: Thomas Kreppel	
Project: UltraZohm_ADC.PrjPCB					